Problem 15.1.16

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty} \frac{(20+30i)^n}{n!}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty} \left|\frac{(20+30i)^{n+1}}{(n+1)!} \cdot \frac{n!}{(20+30i)^n}\right| \\
&= \lim_{n\to\infty} \left|\frac{20+30i}{n+1}\right| \\
&= \lim_{n\to\infty} \frac{\sqrt{1300}}{n+1} \\
&= 0 < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.17

Is the series convergent or divergent? Give a reason.

$$\sum_{n=2}^{\infty} \frac{(-i)^n}{\ln n}$$

Solution

The numerator is bounded cyclically so only the denominator affects the series convergence.

$$\frac{1}{\ln n} > \frac{1}{n} \ \forall n \in \mathbb{Z} \geq 2$$

Therefore, the series diverges by direct comparison to the divergent harmonic series.

Problem 15.1.18

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty} n^2 \left(\frac{i}{4}\right)^n$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(n+1)^2\left(\frac{i}{4}\right)^{n+1}}{n^2\left(\frac{i}{4}\right)^n}\right| \\
&= \lim_{n\to\infty}\left|\frac{(n+1)^2}{n^2}\cdot\frac{i}{4}\right| \\
&= \frac{1}{4}\cdot\lim_{n\to\infty}\frac{(n+1)^2}{n^2} \\
&= \frac{1}{4}\cdot 1 \\
&= \frac{1}{4} < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.19

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty}\frac{i^n}{n^2-i}$$

Solution

$$|a_n| = \left|\frac{i^n}{n^2-i}\right| = \frac{1}{\sqrt{n^4+1}}$$

$$|a_n| < \frac{1}{n^2}\ \forall n \in \mathbb{Z} > 0$$

Therefore, the series converges by direct comparison to the convergent p-series $\sum_{n=0}^{\infty}\frac{1}{n^2}$.

Problem 15.1.20

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty}\frac{n+i}{3n^2+2i}$$

Solution

The real part diverges due to the harmonic series and therefore the series diverges.

$$\sum_{n=0}^{\infty}\frac{n}{3n^2} = \frac{1}{3}\sum_{n=0}^{\infty}\frac{1}{n}$$

Problem 15.1.21

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty}\frac{(\pi+\pi i)^{2n+1}}{(2n+1)!}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(\pi+\pi i)^{2n+3}}{(2n+3)!}\cdot\frac{(2n+1)!}{(\pi+\pi i)^{2n+1}}\right| \\
&= \lim_{n\to\infty}\left|\frac{(\pi+\pi i)^2}{(2n+3)(2n+2)}\right| \\
&= 0 < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.22

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}$$

Solution

$$\frac{1}{\sqrt{n}} > \frac{1}{n}\ \forall n \in \mathbb{Z} > 1$$

Therefore, the series diverges by direct comparison with the harmonic series.

Problem 15.1.23

Is the series convergent or divergent? Give a reason.

$$\sum_{n=0}^{\infty}\frac{(-1)^n(1+i)^{2n}}{(2n)!}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(-1)^{n+1}(1+i)^{2(n+1)}}{(2(n+1))!}\cdot\frac{(2n)!}{(-1)^n(1+i)^{2n}}\right| \\
&= \lim_{n\to\infty}\left|\frac{-(1+i)^2}{(2n+2)(2n+1)}\right| \\
&= 0 < 1, \text{ therefore the series converges}
\end{aligned}$$

Problem 15.1.24

Is the series convergent or divergent? Give a reason.

$$\sum_{n=1}^{\infty}\frac{(3i)^n n!}{n^n}$$

Solution

Ratio Test:

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty}\left|\frac{(3i)^{n+1}(n+1)!}{(n+1)^{n+1}}\cdot\frac{n^n}{(3i)^n n!}\right| \\
&= \lim_{n\to\infty}\left|\frac{(3i)^{n+1}}{(3i)^n}\cdot\frac{(n+1)!}{n!}\cdot\frac{n^n}{(n+1)^{n+1}}\right| \\
&= \lim_{n\to\infty}\left|3i\cdot(n+1)\cdot\frac{n^n}{(n+1)^n(n+1)}\right| \\
&= 3\lim_{n\to\infty}\left|\left(\frac{n}{n+1}\right)^n\right| \\
&= 3\lim_{n\to\infty}\left|\frac{1}{\left(\frac{n+1}{n}\right)^n}\right| \\
&= 3\lim_{n\to\infty}\left|\frac{1}{\left(1+\frac{1}{n}\right)^n}\right| \\
&= 3\cdot\frac{1}{e} \\
&= \frac{3}{e} > 1, \text{ therefore the series diverges}
\end{aligned}$$

Problem 15.2.7

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}\left(z-\frac{1}{2}\right)^{2n}$$

Solution

Problem 15.2.9

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty}\frac{n(n-1)}{3^n}(z-i)^{2n}$$

Solution

Problem 15.2.13

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty}16^n(z+i)^{4n}$$

Solution

Problem 15.2.15

Find the center and the radius of convergence.

$$\sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2}(z-2i)^n$$

Solution

Problem 15.2.17

Find the center and the radius of convergence.

$$\sum_{n=1}^{\infty} \frac{2^n}{n(n+1)} z^{2n+1}$$

Solution

Problem 15.3.5

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=2}^{\infty} \frac{n(n-1)}{2^n}(z-2i)^n$$

Solution

Problem 15.3.7

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{n}{3^n}(z+2i)^{2n}$$

Solution

Problem 15.3.9

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{(-2)^n)}{n(n+1)(n+2)} z^{2n}$$

Solution

Problem 15.3.11

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=1}^{\infty} \frac{3^n n(n+1)}{7^n} (z+2)^{2n}$$

Solution

Problem 15.3.13

Find the radius of convergence in two ways:

(a) Directly by the Cauchy–Hadamard formula in Sec. 15.2
(b) From a series of simpler terms by using Theorem 3 or Theorem 4

$$\sum_{n=0}^{\infty} \left[\binom{n+k}{k} \right]^{-1} z^{n+k}$$

Solution